Week 4

COGS 108 SP 2026

*By: Jason Chen & Jessica Cheng, Spring 2026

DI slides!

https://github.com/shreyamusini/cogs108_sp26_di_slides_shreya

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% git clone https://github.com/shreyamusini/cogs108_sp26_di_slides_shreya.git
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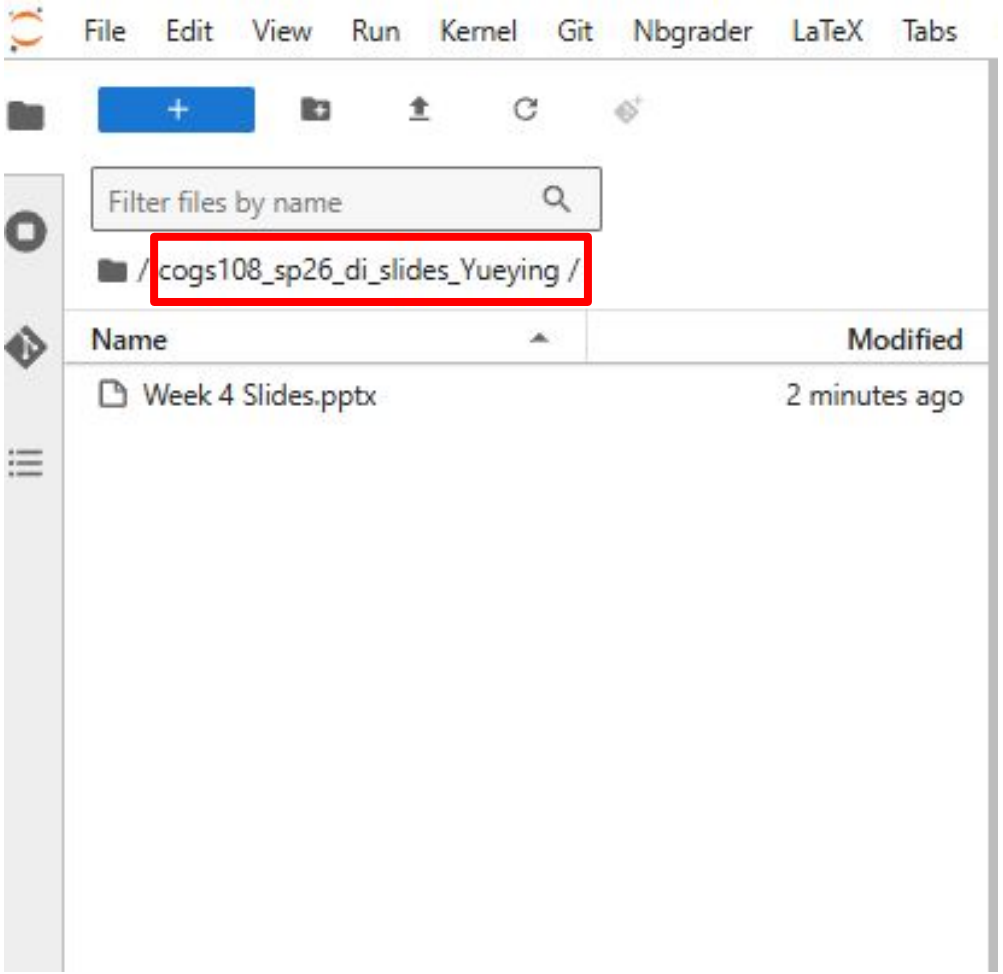
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Week 4 Slides.pptx
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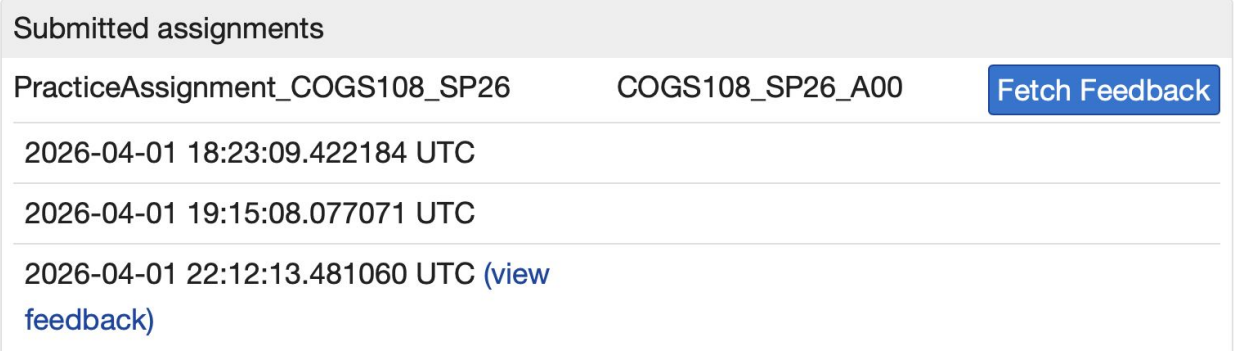
Due dates

- A1 and Project Review: Wednesday (04/22)
- D3: Friday (04/24)

Reminder:
Datahub Feedback



Data Ethics: core ideas

- **Principles**
  - Respect for persons / autonomy
  - Beneficence: aim to help
  - Non-maleficence: avoid harm
  - Justice: fairness in burdens and benefits
- **Reminders**
  - Good intentions do not guarantee ethical outcomes
  - Harms can be indirect, delayed, or unintended
  - Biased data can produce biased results
  - A system can be legal, technical, and still unethical

Data Ethics: core ideas

- Think like a reviewer
  - Who are the stakeholders?
  - Who could be harmed?
  - What assumptions are built into the data?
  - Can people contest or appeal decisions?

Privacy, PII

- Personally identifiable information (PII)
  - **Direct identifiers**: name, SSN, license number
  - **Indirect identifiers** can also matter if data are linkable
  - **Location, education, financial, and medical data can become identifying in context**

Proxies, and consent

- **Proxy variables**
  - A feature may stand in for a sensitive trait without naming it directly
  - That can still create **unfair or discriminatory outcomes**
  - Always ask what the variable is really capturing
- **Informed consent**
  - Participation should be voluntary
  - People should understand what is collected and the risks
  - Consent can be withdrawn in many research settings

Strong data science questions

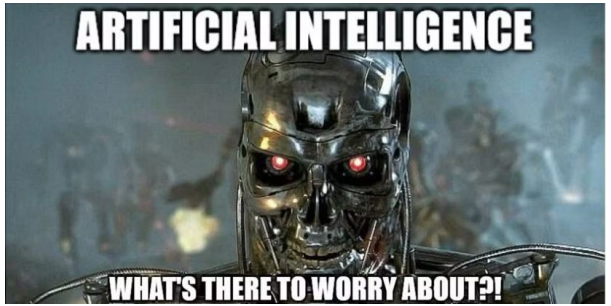
- Specific
- Measurable
- Answerable with obtainable data
- Clear about variables, population, and time frame
- Narrow enough to do, broad enough to matter

Strong data science questions

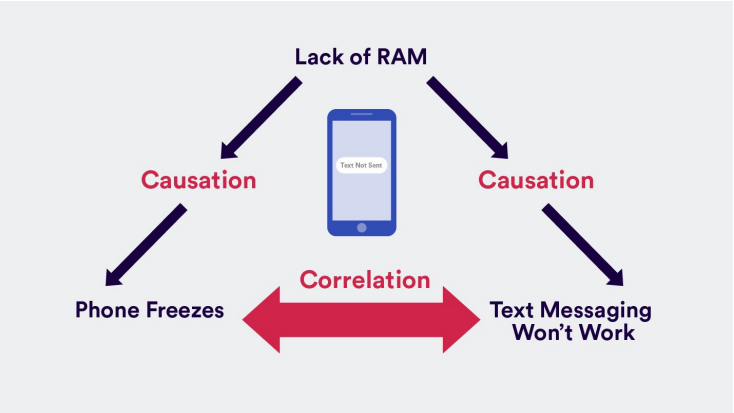
- Red Flags
  - Too vague
  - Undefined outcomes
  - No clear metric
  - Requires unavailable data
  - Asks for causation with only observational data

Evaluation: metrics are not the same as goals

- **Goodhart's law**
  - When a measure becomes a target, it stops being a reliable measure
  - Optimizing the proxy can distort the real goal
- Common failure mode
  - A platform optimizes for clicks, time, or engagement
  - Those metrics may not represent truth, learning, safety, or well-being



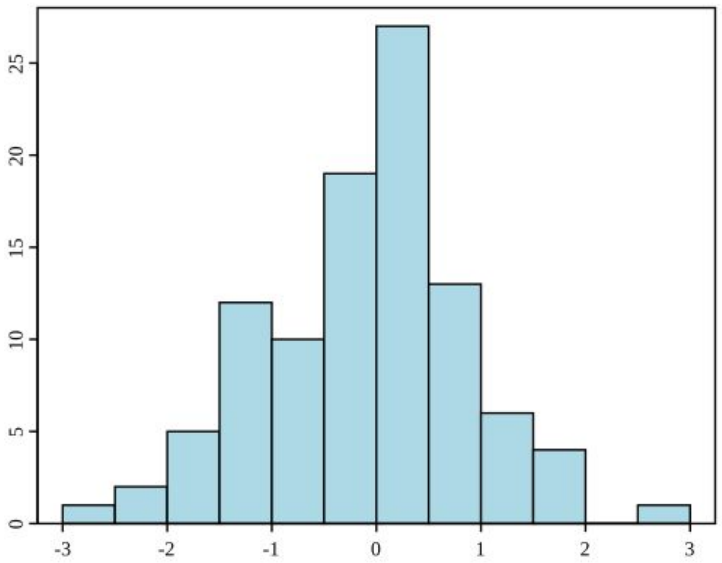
Correlation, causation, and confounding



- **Correlation** tells you
  - Direction and strength of association
  - How variables move together
- Correlation does NOT tell you
  - That one variable **caused** the other
  - That there is no hidden third variable
  - That the relationship is meaningful outside the observed setting

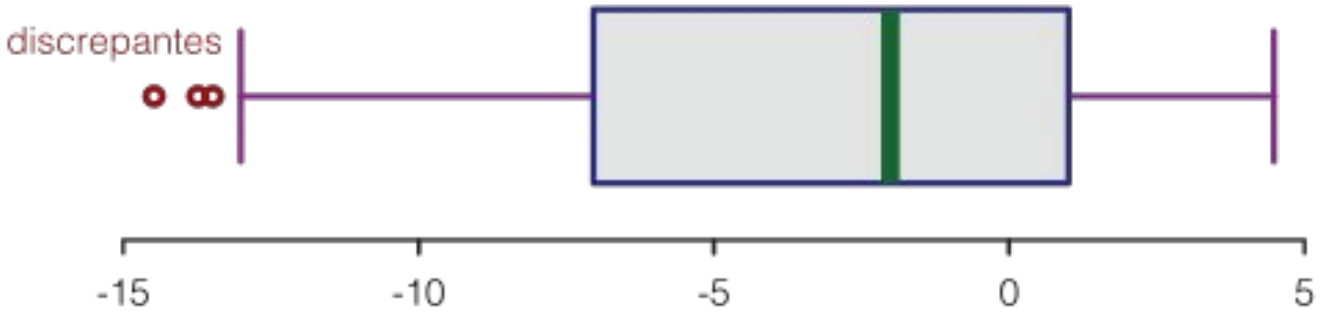
Visualization

- Histogram:
  - one numeric distribution



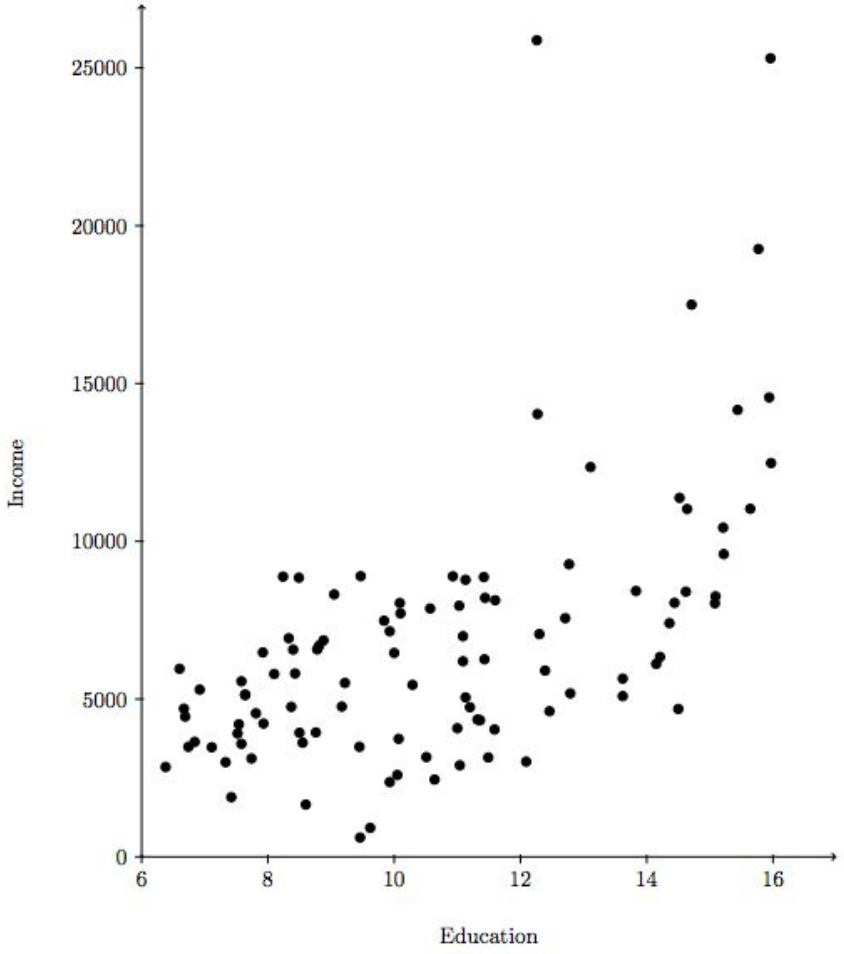
Visualization

- Boxplot:
  - numeric distribution across groups



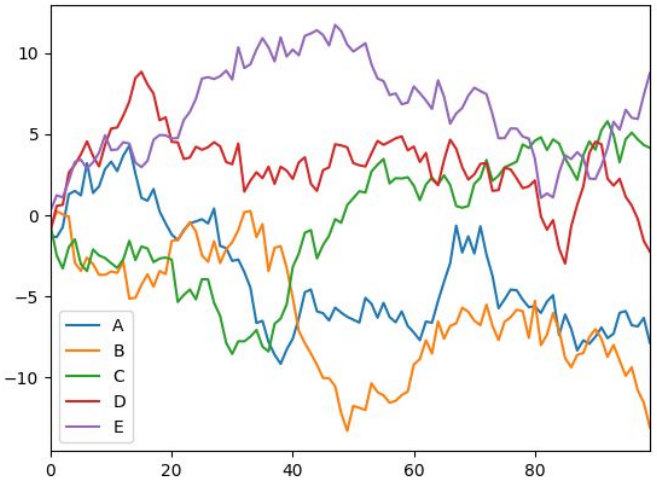
Visualization

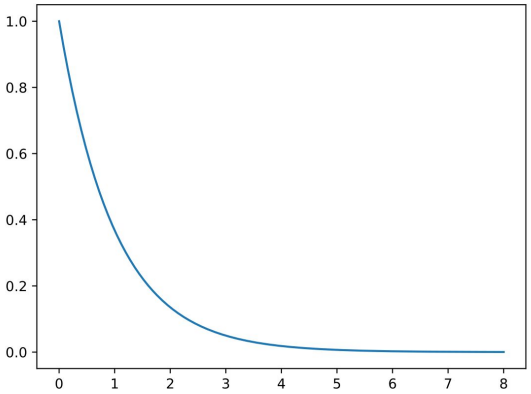
- Scatterplot:
  - relationship between two numeric variables



Visualization

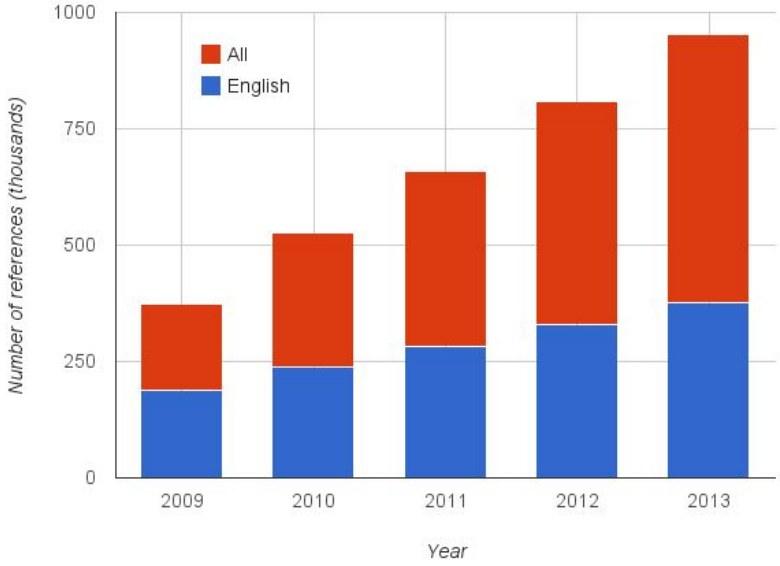
- Line plot:
  - change over time





Visualization

- Stacked bar chart:
  - composition/proportions across categories



Visualization choices

- General design habits
  - Use informative titles and labels
  - Prefer comparisons people can judge accurately
  - Annotate clearly when it reduces confusion
  - Avoid decorative choices that hide the pattern

Library choices

- seaborn
  - high-level statistical graphics
- matplotlib
  - detailed control and fine-tuning

seaborn matplotlib

Common analysis types

- **Descriptive**
  - Summarize what is in the data
  - Counts, means, missingness, tables
- **Exploratory**
  - Look for patterns and generate ideas
  - Flexible and discovery-oriented
- **Inferential**
  - Use sample data to say something about a broader population
  - Often involves uncertainty estimates
- **Predictive**
  - Forecast unknown or future outcomes
  - Accuracy matters more than explanation

Quiz Time

NO Phone, NO Communication, No Notes

Access Code: code here

Your time to …

- Work on D3